

MOUNAH HONORÉ CÉSAIRE

System & Performance Engineer

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SUMMARY

Linux kernel developer and Ph.D. candidate (Jan 2026) specializing in performance optimization and system bottleneck analysis. Proven track record with 5× performance improvements on WireGuard VPN and contributions to kernel networking. Strong expertise in low-level debugging, eBPF, and cloud-native architectures.

WORK EXPERIENCE

Ph.D: Understanding Linux Scheduling Bottlenecks

Inria Rennes

📅 December 2022 – Ongoing 📍 Rennes, France

Kernel VPN Performance Optimization (WireGuard)

- Investigated performance bottlenecks in the WireGuard VPN module, identifying Execution Order Inversion (Eol) as the root cause of a 5× throughput drop.
- Designed and implemented asynchronous kernel APIs, achieving 5× throughput improvement and 2× latency reduction. Findings presented at SYSTOR'25.

Energy Efficiency Analysis of Intel Hybrid CPUs

- Built an automated benchmarking framework to evaluate Intel heterogeneous CPUs (P-cores/E-cores) across DVFS governors and workload types.
- Delivered actionable insights for scheduler design and CPU tuning, balancing performance and energy efficiency.

Research Internship

IRISA

📅 June 2022 – September 2022 📍 Rennes, France

Cloud-Native Malware Detection Platform on Serverless

- Designed and deployed a serverless detection pipeline using OpenFaaS and OpenWhisk, enabling on-demand analysis of Android network traffic.
- Benchmarked and compared FaaS-based detection performance with traditional anti-malware solutions.

PUBLICATIONS

👥 Conference Proceedings

- H. C. Mounah, D. Mvondo, J. Lawall, and Y.-D. Bromberg, “The impact of kernel asynchronous apis on the performance of a kernel vpn,” in *Proceedings of the 18th ACM International Systems and Storage Conference*, ser. SYSTOR '25, Virtual, Israel: Association for Computing Machinery, 2025, pp. 167–173, ISBN: 9798400721199. DOI: 10.1145/3757347.3759133.

EDUCATION

Ph.D in Computer Science

University of Rennes

📅 Dec 2022 – Ongoing

Engineering Degree in Computer Science

National Advanced School of Engineering of Yaounde, Cameroon

📅 Sept 2017 – Oct 2022

TECHNICAL SKILLS

Languages: C, Go, Python, Bash, JavaScript

Kernel & Performance: eBPF, Perf, ftrace, kernel debugging

Cloud & DevOps: Docker, Kubernetes, Ansible, OpenFaaS

CI/CD: GitLab CI, GitHub Actions

Version Control: Git

SOFT SKILLS

Teamwork and Collaboration

Problem Solving

Communication

Critical Thinking

LANGUAGES

- French:** Native
- English:** Fluent (written and spoken)